

UCL SGNL OPS / Orbit Dynamics Library

Assessed from the working tree at ~/odl20lite, x86-64 Linux, GCC 15

Verdict

Seven phases of testing leave a narrow, well-evidenced position. The dynamics are validated to a **3.6 cm median** against IGS final orbits — close to the 2.5 cm accuracy of the reference itself. One estimated parameter recovers a physical property precisely enough to sort the GPS constellation into hardware blocks it was never told about, and that survived every test applied to it. **None of this is a product:** the tool fits orbits that are already known, it does not determine them from observations. The single thesis that could carry a business — characterising high area-to-mass objects — is demonstrated in simulation and untested on a real object. Obtaining one is now the only question that matters.

Closing numbers

Adding a five-term empirical radiation-pressure set to the estimated state:

across 30 GPS satellites	7 parameters	with ECOM
median fit residual	0.142 m	0.036 m
worst case	0.490 m	0.044 m
spread across satellites	wide	every satellite 2.5–4.4 cm

This is where the measurement stops being about the software. IGS finals are quoted at about 2.5 cm, so a 3.6 cm residual is approaching the accuracy of the truth data itself. Existing seven-parameter results were verified bit-identical after the refactor before any of this was trusted.

What that number is not

It is a *fit residual to a known trajectory*, not an orbit determination accuracy. It says the force model can reproduce a precise orbit to 3.6 cm over a day. It does not say orbits can be determined to 3.6 cm, because that requires ingesting real tracking observations — and this software has no observation models at all. The distinction is the difference between validating dynamics and operating a service, and it is why the valuation below is unchanged by an otherwise excellent number.

The hardware-block finding, final form

A single estimated effective AC_R/m sorts 31 GPS satellites into the five hardware blocks recorded in the IGS antenna file, with no misclassifications and without being told blocks exist. Every test applied to it has strengthened it:

- **The confound was measured**, not argued away — real within blocks, and an order of magnitude too small to explain the separation.
- **A cross-check absorbed error has no reason to pass:** multiplying each block's estimate by its published on-orbit mass — never seen by the fit — puts three GPS IIR sub-blocks within 4% of a common effective area. Those three share a bus, which the fit was also not told.
- **It reproduces across dates** to 0–0.5%, including two satellites returning identical estimates to four significant figures on independent arcs *at two different values*.
- **It survives five parameters.** Block separation is marginally sharper with ECOM than without,

every block mean moving under 0.5%.

A caveat reported in revision 4 has been withdrawn, and the finding is stronger than stated then. The GPS III block's 6.5% internal scatter was flagged as possibly absorbed model error. ECOM removes 92% of that block's residual — 0.286 m to 0.030 m — and the scatter does not move, staying at 6.43% with the block mean shifting 0.4%. Had the scatter been radiation-pressure-shaped error soaking into the parameter, removing the error would have moved it. The estimate is robust to a fourfold reduction in residual. The spread is real variation between five vehicles and remains unexplained — a much smaller caveat than the one it replaces.

The open question, and what bounds it

The estimator's fractional precision scales exactly as $1/\text{signal}$, measured across a thousandfold synthesised span. GNSS sits at $0.021 \text{ m}^2/\text{kg}$ and fails the separation threshold at realistic tracking noise; the high area-to-mass property sets in this source span 0.16 to 6.3 and clear it by 5 to 190 times. Tumbling does not break this — fast variation integrates out, slow variation biases 3.3% while moving the residual by three orders of magnitude, so it is flagged rather than silent.

Three limits on that, stated rather than left to be found

The floor is measured for attitude and unmeasured for shape. Precision improves as $1/\text{signal}$ only down to a floor set by how well the object's shape and attitude are known, and that floor does not improve — confirmed at 3.28%, 3.28%, 3.27% across a hundredfold signal range. But in a cannonball model, reflectivity and area enter as one scalar that is perfectly degenerate with the estimated parameter, so a wrong reflectivity is absorbed exactly and produces no residual at all. *This tool cannot detect shape error, and shape is unknown by definition for the target population.*

There is an instrument for that already in the tree, and it is not the missing ray-tracer. `Force_rp_box_wing.cpp` is present, selectable as `rp_model = 3`, and depends on none of the absent grid files — it needs per-face areas and optical properties, which nine named spacecraft in the source already carry, including a purpose-built `testbox`. It was not used anywhere in this work, and it is the natural next instrument for shape sensitivity.

Two traps sit in front of anyone who follows that pointer, and both were found by following it:

- **It is silently disabled unless a flux model is also on.** `Configuration.cpp` zeroes `rp_model` whenever `srp` and `erp` are both off — which every shipped configuration has. Setting `rp_model = 3` alone produces a clean run, well-formed output, and a trajectory *byte-identical* to no model at all. The logic is right, since a radiation pressure model with no flux has nothing to act on, but the override is silent and the dependency is undocumented. With `srp` enabled the model works and produces a physically sized effect — 0.35 m over 45 minutes at GEO for GPS-IIR, against an otherwise identical run with no shape model.
- **It needs a spacecraft that actually defines the properties.** The default test object in every shipped config is not a named entry in the source at all, so it falls through to defaults. A short propagation tolerates this; a 22-hour orbit fit diverges.

Everything about high area-to-mass objects here is synthetic, with truth arcs generated by the same force model that then fitted them — a fair test of the estimator and an unfair test of the physics. Nothing has been tested against an object whose shape and attitude are genuinely unknown. For objects in low orbit, drag and radiation pressure are also near-degenerate, both scaling with area over mass; largely moot in geostationary orbit, where the debris of interest mostly sits.

What this engagement produced

- A tree that did not build, now under version control with documented provenance and a licence inventory.
- An honest accuracy figure against public reference data, reproducible by one script.
- A characterised estimator with measured scaling behaviour and a known failure boundary.
- A publishable finding: spacecraft hardware classes are recoverable from dynamics alone.
- **Five silent-wrong-answer defects found and fixed.** A least squares built on identity partials; a header-line count that restricted all validation to one constellation; an assumed sampling interval; observations matched to states by index; and a configuration substitution that failed without complaint. Two further silent failures were found and are described above: a non-finite epoch passing a range check because every comparison against NaN is false, and a radiation-pressure model silently disabled by an unrelated switch. *This codebase's failure mode is consistently a plausible result rather than an error, and that is the single most important thing for anyone inheriting it to know.* It is not a historical observation: the last two instances were found on the final day, one of them by following a pointer in an earlier draft of this document.

Valuation

- **As an orbit determination product: approximately £0**, unchanged across all eight revisions, and not altered by the 3.6 cm figure — a tool that cannot ingest an observation cannot determine an orbit.
- **As a high area-to-mass characterisation capability: unproven, and the only thesis that could carry a business.** One real object away from being testable.
- **As a validated body of work: the durable result**, and the one that survives whichever way the open question goes.
- **Replacement cost remains the wrong frame** at £300k–£600k.

Recommendation

The engineering is finished and should stop here. What remains is not a development question.

1. **Obtain one real high area-to-mass object with a precise ephemeris.** This is the whole question. It is a data-access problem — currently blocked on a credentialed download — and it is worth more than any further work on the code.
2. **If it confirms the synthetic result:** the remaining investment is observation models and covariance, and the box-wing model above is the cheap first step for shape sensitivity. Do not start either before the IP position is settled.
3. **If it does not, or the data cannot be obtained: stop and publish.** Recovering spacecraft hardware classes from dynamics alone is an interesting result on its own terms. Finishing there is a legitimate outcome, not a failure.
4. **Settle the IP position regardless.** Outstanding since revision 1. It gates publication as surely as sale and needs no engineering at all.

Basis: measured facts come from the tree at `/home/rog/odl20lite` (final commit `a710c5a`, clean tree, all checks and baselines passing, verified here) and from runs executed on 15 September 2026 by the `odl maintainer` session, reviewed and spot-verified. High area-to-mass results are synthetic, as described. Competitive and valuation sections are judgement. Revision 9 is final: it restores a GPS III figure withdrawn in revision 4, whose caveat has since been disproved, and corrects the box-wing note in revision 8, which was accurate about the model and silent about two conditions required to run it.